

FULL NAME: _____

BLOCK: _____

Rational Expressions — Equivalent & Simplify

Lesson 4-3 · Period 1 · Student Packet

OBJECTIVES

Math Objective	Write equivalent rational expressions over a restricted domain, identify domain restrictions from factored denominators, and simplify rational expressions by canceling common factors.
Language Objective	Explain, in writing and orally using sentence frames, why a value is excluded from the domain and why we cancel factors but not terms.
Essential Understanding	A rational expression equals an equivalent simpler form ONLY when both share the same domain — canceling a factor never erases the restriction it imposed.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Write equivalent rational expressions, simplify, and identify domain restrictions from polynomial denominators.
Essential Question	Why does canceling a common factor not remove its domain restriction?
Materials	Student packet, pencil, highlighter. Teacher displays worked Example 1 then gradual-release into Example 2.

LAUNCH — Rational Expressions (teacher models on screen)

[BOOK] EQUIVALENT + SIMPLIFY RULES (keep printed on your page)

1. Factor the numerator AND denominator completely.
2. Cancel only common **FACTORS** (never terms across + or -).
3. State domain restrictions: every zero of the **ORIGINAL** denominator is excluded — including factors you canceled.

[WARN] COMMON ERROR

Do **NOT** cancel terms that are added or subtracted across a fraction bar.
 You can only cancel factors that multiply across the entire numerator or denominator.
 Test: factor completely **FIRST**, then cancel only identical factor pairs.

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Use the Equivalent + Simplify Rules above for every problem. Move through items in order — each builds on the last.

Bank-item labels (in order):

1. Try It 1 — Write an equivalent rational expression
2. Practice #20 — Identify equivalent expressions

3. Practice #22 — Identify domain restrictions
4. Practice #24 — Apply domain restrictions
5. Try It 2 — Simplify by factoring and canceling
6. Practice #17 — Simplify
7. Practice #19 — Simplify with domain restrictions (extension)

Try It 1 — Write an equivalent rational expression

Write an expression equivalent to $(x - 4)/(x)$ over the domain $\{x \mid x \neq 0 \text{ or } -2\}$.

Practice #20 — Identify equivalent expressions

Write an equivalent expression over the given domain.

$(x(x - 2))/(x - 5)$ for all x except 5 and -6

Practice #22 — Identify domain restrictions

What is the simplified form of the rational expression? What is the domain?

$(y^2 - 5y - 24)/(y^2 + 3y)$

Practice #24 — Apply domain restrictions

What is the simplified form of the rational expression? What is the domain?

$(x^2 + 8x + 15)/(x^2 - x - 12)$

Try It 2 — Simplify by factoring and canceling

Simplify each expression and state the domain.

a. $(x^2 + 2x + 1)/(x^3 - 2x^2 - 3x)$

b. $(x^3 + 4x^2 - x - 4)/(x^2 + 3x - 4)$

Practice #17 — Simplify

Construct Arguments Explain how you can tell whether a rational expression is in simplest form.

Practice #19 — Simplify with domain restrictions (extension)

Reason If the denominator of a rational expression is $x^3 + 3x^2 - 10x$, what value(s) must be restricted from the domain for x ?

SHARE / SUMMARY (5 min)

Sentence frames

Pair share: “The factored form is ____, so the domain excludes $x = \underline{\hspace{1cm}}$. After canceling the factor ____, the simplified expression is ____ with the same domain restriction.”

Whole class: one pair reads their sentence. Class signals thumbs up/sideways.

To simplify $(x^2 + 3x - 10)/(x^2 - 4)$ I first factor to ____.

The restricted domain values are $x \neq$ ____.

After canceling the factor ____, my simplified expression is ____.

One biggest thing I learned today: _____

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early or want extra practice before Period 2.

[SCOPE] OPTIONAL • Model & Discuss

Return to the Do Now graph. With your partner:

B) Use Structure. Why can the graph never have a point at the skipped x-value, no matter what simplified form we write?

C) Examine at least two rational expressions you simplified in Explore. For each, explain in your own words why the domain restriction from the canceled factor still applies.

Optional #1 (Savvas Practice)

What is the simplified form of the rational expression? What is the domain?

$$(x^3 + 9x^2 - 10x)/(x^3 - 9x^2 - 10x)$$